

RG-01

// AUTONOMOUS_DRONE



RINGIL

aeluin::technologies

[+] MISSION

SURVEILLANCE / RECON

[+] PERFORMANCE

20 MIN AUTONOMY
80 KM/H VMAX

[+] COMMUNICATION

ISM 433 MHZ (LoRa)
FHSS-LoRa HOPPING

[+] WEIGHT

< 1.5 KG

1.0 SYSTEM ARCHITECTURE OVERVIEW

The RG-01 is a high-performance uncrewed hexacopter platform designed for multi-agent autonomous swarm deployment. Engineered to operate reliably in challenging, GPS-denied environments, the platform relies on a secure, real-time operating system. By processing data directly on-device, the RG-01 eliminates reliance on centralized communication, reducing latency and preventing single points of failure during critical missions.

Cruise Velocity	55 km/h
Max Intercept Velocity	80 km/h
Operational Range	12 km
Endurance (Full Load)	20 min
Pre-flight Deployment	< 45 s
Total Structural Mass	< 1.5 kg
Onboard Compute	NVIDIA Jetson
RF Band Allocation	ISM 433 MHz (LoRa)

3.0 GPS-DENIED NAVIGATION & ONBOARD MAPPING

To guarantee continuous operation in environments where GPS signal is blocked or lost, the RG-01 utilizes an advanced intelligent navigation engine. By combining data from built-in internal motion sensors with real-time stereo-vision processing, the drone achieves highly accurate tracking without external aids. This continuous self-localization loop allows the platform to know its exact position and map its surroundings.

[+] CORE FUNCTIONAL INTEGRATION

- Decentralized Mesh Networking** Smart drone-to-drone communication without central command overhead.
- Interference-Resistant Links** Protected signal frequencies built to withstand tactical signal jamming.
- Sensor-Based Return-to-Base** Automatic, secure routing back to home base using internal backup sensors.
- Stabilized Dual-Camera Array** Integrated optical and infrared sensors for automated target tracking.

2.0 AERODYNAMIC STRUCTURE & SIGNATURE

Built from advanced, lightweight carbon-fiber composites, the drones streamlined airframe is optimized for low operational visibility, significantly reducing its acoustic and radar signatures. This durable, rugged construction ensures steady multi-axis flight stability, allowing the platform to maintain position and perform precise maneuvers even under severe wind loads and harsh weather conditions.

4.0 OPERATIONAL TOPOLOGY & SWARM NETWORKING

The platform features integrated communication arrays that allow multiple drones to coordinate and distribute flight paths entirely on their own. Swarm units constantly share tracking data with one another. This enables automated formation control, area coverage, and intelligent task allocation without requiring real-time manual intervention from human operators.

5.0 PERCEPTION & REAL-TIME AVOIDANCE

The RG-01 constantly captures environment depth data using on-board high-frame-rate cameras. This sensory data is processed instantly on the drone to build a live, 3D digital model of the surrounding airspace.

Instead of basic proximity alerts that just warn of close objects, the flight engine calculates optimal, smooth path adjustments in real time. The navigation system actively avoids unexpected obstacles, moving terrain, safe zones, and neighboring drones within the same swarm.

[+] AUTONOMY STACK COMPONENT MAPPING

Software Layer	Primary Inputs	Operational Outcome
Navigation Engine	Optical + Sensors	Continuous tracking without GPS.
Onboard Mapping	Live Cam + Depth	Real-time 3D airspace maps.
Flight Planner	Maps + Swarm Mesh	Collision-free path optimization.

This document outlines proprietary technical specifications intended exclusively for vetted investment evaluation. Asset protection and hardware parameters are strictly managed via internal compliance agreements.